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Structural characterization of telechelic polyisobutylene diol

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ABSTRACT

The chemical homogeneity of telechelic polyisobutylene diol (PIB-diol), prepared by hydroboration-oxidation of allyl telechelic PIB obtained by reacting living PIB with allyltrimethylsilane, was investigated by liquid chromatography at critical conditions (LCCC) and HPLC coupled with matrix-assisted laser desorption/ionization time-of-flight mass spectrometry (MALDI-TOF MS). A normal phase gradient HPLC method was developed that was able to separate the as-synthesized PIB-diol into three components; PIB-diol, PIB-monool and PIB without any —OH functionality. These were analyzed by MALDI-TOF MS, which suggested that the reaction of living PIB with allyltrimethylsilane was incomplete. LCCC using refractive index (RI) detector as a concentration detector allowed separation and quantification of PIB species according to their chemical heterogeneity (PIB-diol = 95.3%, PIB-monool = 3.3%, non-functional PIB = 1.4%). The calculated number average functionality (F_n) of PIB-diol = 1.94 suggests high quality of PIB-diol suitable for high molecular weight polyurethane synthesis.

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1. Introduction

Thermoplastic polyurethanes (TPUs) are segmented multiblock copolymers containing alternating soft segment (SS) and hard segment (HS). Due to their biocompatibility they have been extensively employed as biomaterials. Most commercially available TPUs use low molecular weight (MW) polyester or polyether diols as the SS. However, these TPUs are unstable against hydrolytic or oxidative degradations, which limit their *in vivo* lifetime. To overcome this problem, we recently introduced polyisobutylene (PIB) based TPUs, which exhibited enhanced biostability against oxidative degradation. [1] The precursor PIB-diol can be prepared by living cationic polymerization followed by end-functionalization. One well known method shown in Fig. 1 employs hydroboration–oxidation of allyl-telechelic PIB obtained by reacting living PIB with allyltrimethylsilane. For the synthesis of TPUs, the number average functionality (F_n) of PIB-diol should be close to the ideal value of 2.0, since the number average degree of extension and thus the ultimate number average molecular weight (M_n) of the TPU is very sensitive to F_n. [2] To obtain high molecular weight is highly desirable since the ultimate tensile properties depend very strongly on molecular weight. [3] Due to incomplete functionalization and/or side reactions, some monofunctional and non-functional PIB may form along with PIB-diol (Fig. 1). [4] Thus, it is necessary to confirm the chemical homogeneity of PIB-diol before using it for the synthesis of TPU.

Normal characterization methods, such as NMR and FT-IR spectroscopy are generally unsuitable to determine the existence of a small amount of undesired side product(s). Again, these techniques cannot distinguish polymer chains with different functionalities. Previous researchers have tried to utilize size exclusion chromatography (SEC) for the determination of chemical heterogeneity of polymers [5] and copolymers. [6] However, SEC can only provide limited information on the chemical composition of a polymer, if only interfaced with NMR [7] or mass spectrometry. [8] Subsequently, liquid adsorption chromatography (LAC) in a gradient mode was developed. [9] This technique, when applied to separate polymers in terms of their chemical heterogeneity, often suffers from complications arising from the MW dependency of polymer retention and/or precipitation/dissolution of the polymer. [10]

Liquid chromatography at critical conditions of adsorption (LCCC), established by Belinki et al. [11] and Entelis et al., [12] is

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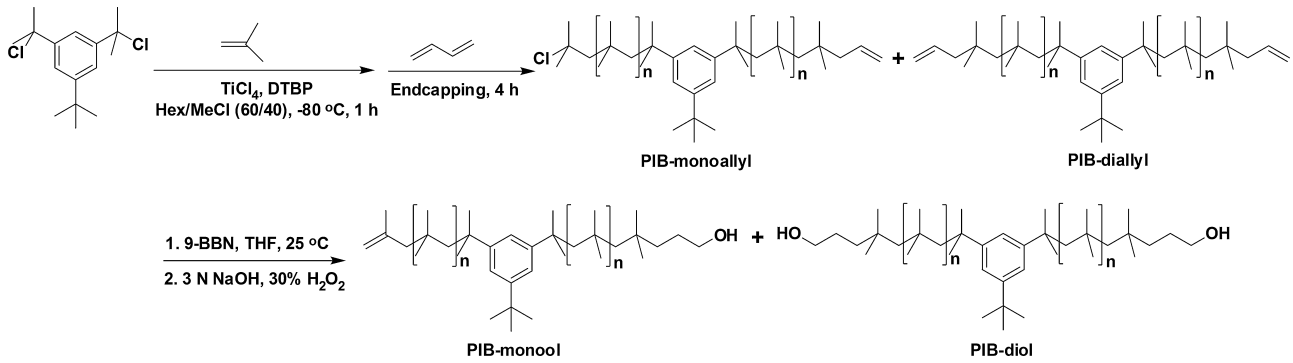


Fig. 1. Synthesis of telechelic PIB-diol.

capable of separating polymers according to their chemical functionalities (such as the number and types of functional groups and their positions in the polymer chain), independently of their MW and determine the chemical heterogeneities. [13] LCCC conditions are in between the condition of SEC, where separation is based on the entropic differences between the macromolecules according to their hydrodynamic volumes and LAC, where separation is based on the enthalpic differences between the macromolecules. [14] At this critical condition, there is a balance between the entropic and enthalpic interactions leading to the exclusive interaction of a single functional group (e.g., end group of the polymer chain) with the stationary phase and separation of macromolecules according to the end group functionality, independently of their MW. [15] Critical conditions are strongly related to the polymer under consideration and thus must be individually determined for every different polymer. [16] There are two practical limitations of LCCC: (a) The development of LCCC method is time-consuming to find the proper eluent composition to reach at the LCCC condition. (b) The most important limitation of LCCC of polymers is reduced sample recovery, especially in case of high molecular weight polymers. This phenomenon is well documented in the literature devoted to LCCC method development. [17] This incomplete polymer elution can sufficiently reduce both precision and accuracy of the quantitative LCCC analyses [18] and may cause a change in the column retentivity in the course of large series of LCCC measurements. [19]

Matrix-assisted laser desorption/ionization-time-of-flight mass spectrometry (MALDI-TOF MS) provides valuable information on the polymer end group structure. [20] The information obtained from MALDI-TOF MS analysis can be a strong supplementary tool in combination with LCCC for characterization of a polymer. [21]

In the present contribution, we focus on developing a LCCC method using RI detector as a concentration detector for the determination of chemical heterogeneity and analysis by gradient HPLC and MALDI-TOF MS to identify the structure of monofunctional and non-functional PIB in the synthesized PIB-diol sample. PIB-diol, an important precursor for TPU synthesis, is not easy to characterize by the described technologies. It has been shown, that LC/MALDI-MS coupling is a powerful combination for polymer heterogeneity characterization and thus synthesis monitoring.

2. Experimental

2.1. Characterization

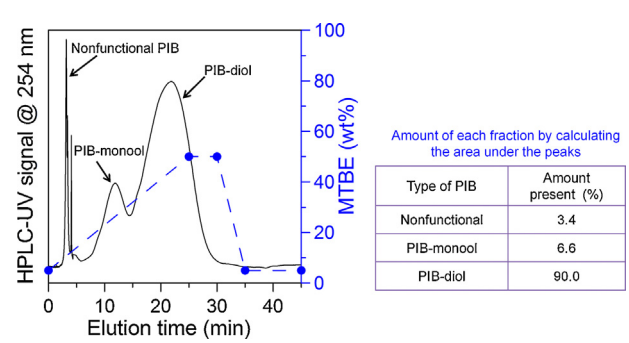
2.1.1. Analysis by gradient HPLC using UV detector

HPLC was conducted on a Waters HPLC system with Waters 2695 separation module at a flow rate of 0.5 mL/min using a Waters 2996 Photodiode Array detector operating at $\lambda = 254$ nm. 10 μL samples of 0.8 wt% polymer solutions were injected. All measurements were carried out at a constant column temperature of 35 °C and constant sample temperature of 30 °C. One normal phase HPLC column (YMC-Pack SIL) with average particle size = 5 μm, pore diameters = 120 Å and column size (Inner diameter *x* Length) = 4.6 mm × 250 mm was used. HPLC grade heptane and methyl tert-butyl ether (MTBE) were used to prepare the mobile phase and the solution of the polymers. The initial solvent composition used was heptane/MTBE 95/5 (*w*/*w*). Solvent composition changes with time (*t*) as follows: heptane/MTBE 95/5 at *t* = 0, heptane/MTBE 50/50 at *t* = 25 min, heptane/MTBE 5/95 at *t* = 30 min and heptane/MTBE 95/5 at *t* = 35–45 min.

2.1.2. Analysis by gradient HPLC and MALDI-TOF MS

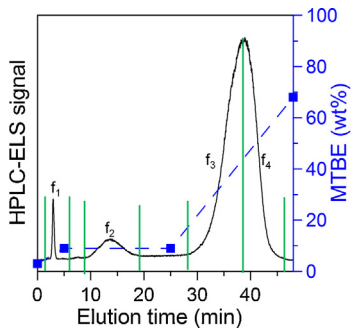
HPLC was conducted on a diol bonded silica column (Nucleosil) with average particle size = 7 μm, pore diameters = 100 Å and column size (Inner diameter × Length) = 2.1 mm × 150 mm at a flow rate of 0.5 mL/min. An evaporative light scattering (ELS) detector (Polymer Laboratories, PL-ELS 2100) operating at an evaporator temperature of 56 °C, nebulizer temperature of 40 °C with a gas flow rate of 1.1 mL/min was used along for the analysis. Then 100 μL samples of 0.45 mg/mL polymer solutions were injected for the analysis. All measurements were carried out at a constant column temperature of 25 °C. HPLC grade heptane and MTBE were used to prepare the mobile phase and the solution of the polymers. The initial solvent composition used was heptane/MTBE 97/3 (*w*/*w*). Solvent composition changes with *t* as follows: heptane/MTBE 97/3 at *t* = 0, heptane/MTBE 91/9 at *t* = 5 min, heptane/MTBE 91/9 at *t* = 25 min and heptane/MTBE 32/68 at *t* = 48 min.

Gradient HPLC and MALDI-TOF MS analysis were done separately. The HPLC fractions were collected and subjected to MALDI-TOF MS analysis. MALDI-TOF MS experiments were performed with a Bruker Reflex III mass spectrometer equipped with a nitrogen laser ($\lambda = 337$ nm), a pulsed ion extraction and a reflector. The MALDI-TOF MS instrument was calibrated with polystyrenes (M_n = 3 kg/mol and 5 kg/mol) with narrow molecular weight



Type of PIB	Amount present (%)
Nonfunctional	3.4
PIB-monool	6.6
PIB-diol	90.0

Fig. 2. HPLC chromatogram of PIBOL014 under gradient condition using UV detector and heptane/MTBE as the eluent. We assumed that PIB-diol, PIB-monool and non-functional PIB have same molar absorption coefficient at 254 nm.



Amount of each fraction by calculating the area under the peaks

Type of PIB	Amount present (%)
Nonfunctional (f_1)	1.62
PIB-monool (f_2)	6.85
PIB-diol ($f_3 + f_4$)	91.53

Fig. 3. HPLC chromatogram of PIBOL014 under gradient condition using ELS detector and heptane/MTBE as the eluent.

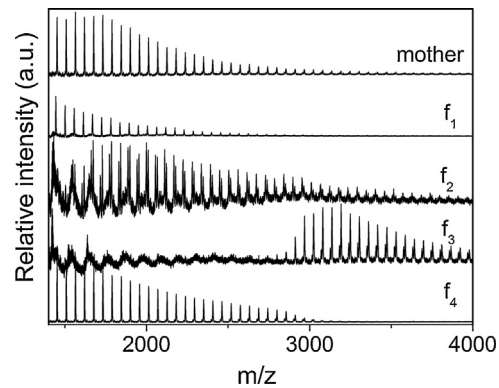


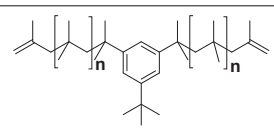
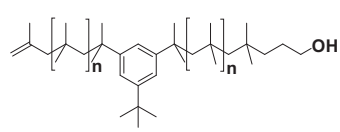
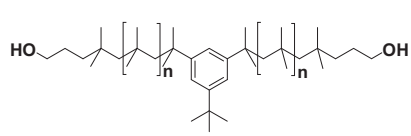
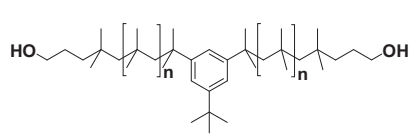
Fig. 4. MALDI-TOF MS spectra of PIBOL014 using dithranol as the matrix and silver trifluoroacetate as the salt.

distribution and well-defined end groups. Dithranol (20 mg/mL in THF) and silver trifluoroacetate (1 mg/mL in THF) were used as the matrix and the salt, respectively. The concentration of the polymer was 5 mg/mL in THF. For measurement, final samples were prepared by mixing a 5 μL aliquot of the polymer solution with 50 μL of the matrix solution and 1.5 μL of salt solution. Typically, 0.5 μL of the mixture (sample/matrix/salt) solution was spotted on a sample plate and allowed to dry in air at room temperature.

2.1.3. Liquid chromatography under critical conditions (LCCC)

LCCC was conducted on a Waters HPLC system with Waters 2695 separation module at a flow rate of 0.5 mL/min. For UV/ELS system, an ELS detector (Varian 385-LC) operating at 50 °C with a gas flow rate of 2.5 L/min was used along with a Waters 2996 Photodiode Array detector operating at λ = 254 nm. For UV/RI system, a Waters 2414 Refractive Index detector at 30 °C was used along with a Waters 2996 Photodiode Array detector operating at λ = 254 nm. 10 μL samples of 0.05 wt% polymer solutions were injected for UV/ELS system, while 10 μL samples of 0.5 wt% polymer solutions were injected for UV/RI system. All measurements were carried out at a constant column temperature of 35 °C and constant sample temperature of 30 °C. One reversed phase C18 chromatography column (YMC HPLC COLUMN) with average particle size = 5 μm, pore diameters = 120 Å and column size (Inner

Table 1
Structural assignment of observed peaks of the PIB fractions peaks shown in Fig. 4.

Fraction	m/z theo.	m/z Exp.	DP	Structure	Type of PIB
[a]f_1	1837.121	1838.069	27		Non-functional PIB
	1893.228	1894.183	28		
	1949.336	1949.222	29		
[b]f_2	2065.539	2066.584	30		PIB-monool
	2121.647	2122.639	31		
	2177.754	2178.729	32		
[c]f_3	3079.463	3080.751	47		PIB-diol (high MW)
	3135.570	3136.809	48		
	3191.678	3192.832	49		
[d]f_4	1845.098	1845.102	25		PIB-diol (low MW)
	1901.205	1901.151	26		
	1957.313	1957.248	27		

[a] Theoretical MW for f_1 = (MW of initiator) + [(MW of IB) × DP] + (MW of silver ion) − [(MW of 'H' × 2] = (216.3666) + (56.1075 × 28) + 107.8677 − (1.0079 × 2) = 1893.228.
[b] Theoretical MW for f_2 = (MW of initiator) + (MW of terminator) + [(MW of IB) × DP] + (MW of silver ion) − (MW of 'H') = (216.3666) + (59.0879) + (56.1075 × 31) + 107.8677 − 1.0079 = 2121.647.
[c] Theoretical MW for f_3 = (MW of initiator) + [(MW of terminator) × 2] + [(MW of IB) × DP] + (MW of silver ion) = (216.3666) + (59.0879 × 2) + (56.1075 × 48) + 107.8677 = 3135.570.
[d] Theoretical MW for f_4 = (MW of initiator) + [(MW of terminator) × 2] + [(MW of IB) × DP] + (MW of silver ion) = (216.3666) + (59.0879 × 2) + (56.1075 × 25) + 107.8677 = 1845.098.

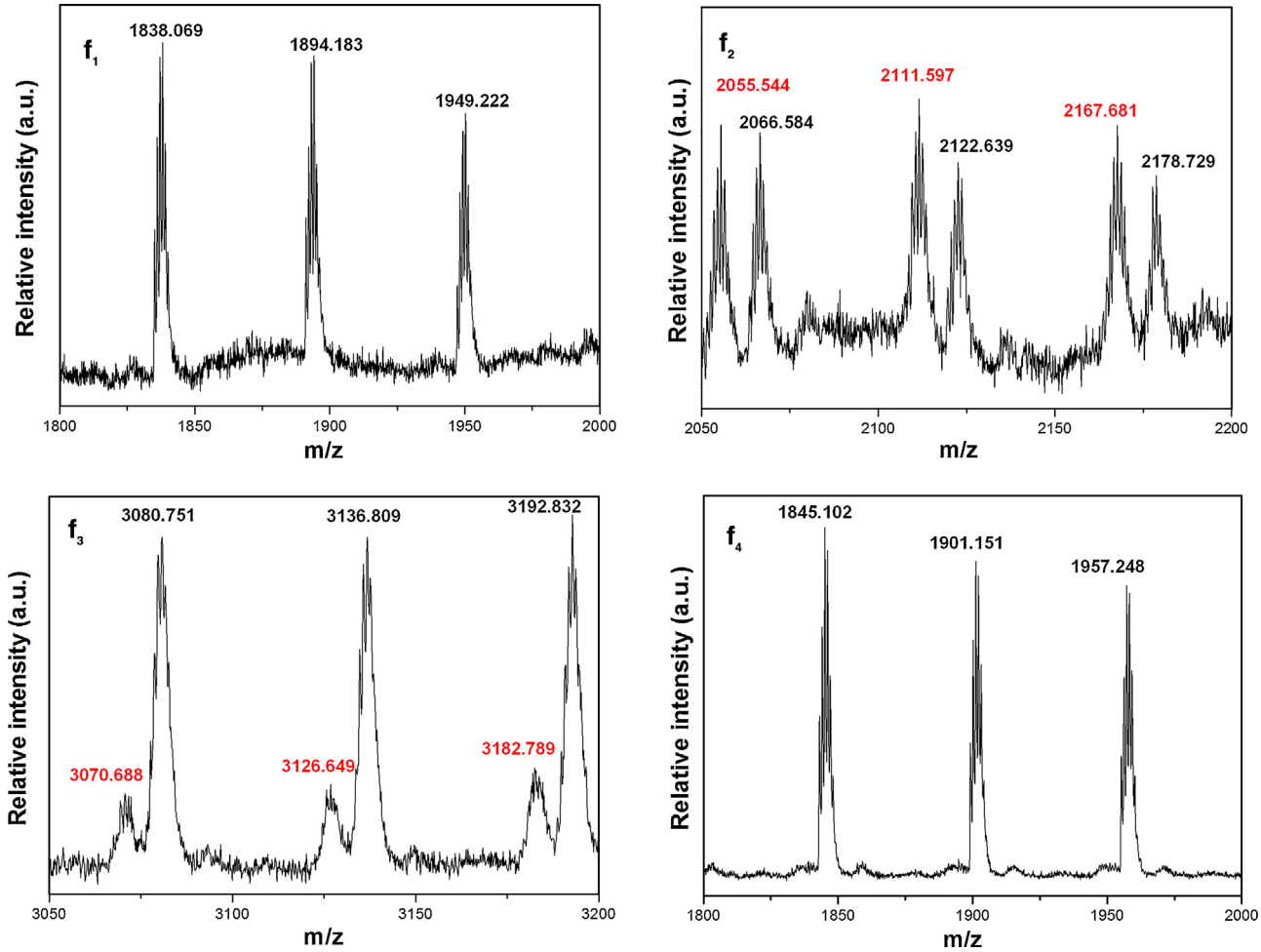


Fig. 5. Enlarged view of MALDI-TOF MS spectra of different fractions (the values in red indicated the unidentified series) using dithranol as the matrix and silver trifluoroacetate as the salt.

diameter × Length) = 4.6 mm × 250 mm was used. HPLC grade THF and methanol were used to prepare the mobile phase and the solution of the polymers. The solvents were freshly distilled before use to avoid any degassing related issue. The critical solvent composition for PIB was established as THF/methanol 80.5/19.5 (*w*/*w*). Premixing of the mobile phase by weight is necessary for a constant and exact composition. Samples were prepared by dissolving polymer in THF/methanol 80.5/19.5 (*w*/*w*).

3. Results and discussion

3.1. Analysis by gradient HPLC using UV detector

PIBOL014 was first analyzed by a solvent gradient HPLC method based on UV detector using heptane/MTBE as the eluent. Non-functional PIB eluted first, well separated from PIB-monool, which was followed by PIB-diol (Fig. 2). The non-functional PIB eluted first as it has no interaction with the column. PIB-monool eluted before PIB-diol as it has less interaction with the column in comparison with PIB-diol species with two hydroxyl groups. However, the separation between the PIB-monool and diol was less clear as high MW (to be explained later) PIB-diol eluted with PIB-monool. The reason why we used ELSD for subsequent analysis instead of UV has nothing to do with the concern of different absorptivity of the PIB species present in PIBOL014. It was simply because UV absorptivity of the PIB sample was too low and ELSD provides a much higher sensitivity.

3.2. Analysis by gradient HPLC using ELSD and MALDI-TOF MS

In a modified solvent gradient method with ELS detection using heptane/MTBE as the eluent an isocratic flow region of heptane/MTBE 91/9 (*w*/*w*) from $t = 5$ min to $t = 25$ min was inserted between two gradient regions. HPLC chromatogram of PIBOL014 (Fig. 3) under this condition revealed three well separated peaks, which correspond to non-functional PIB, PIB-monool and PIB-diol, with PIB-diol as the major component. The amount of each PIB fraction was calculated by calculating the area under the peak, as shown in Fig. 3.

To identify the end group structures, the four fractions i.e., non-functional PIB (f_1), PIB-monool (f_2) and two fractions of PIB-diol (f_3, f_4) indicated on Fig. 3 were analyzed by MALDI-TOF MS (Fig. 4). Fig. 5 shows the expanded MALDI-TOF MS spectra of the four fractions. The enlarged view of a portion of the original spectrum of f_1 shows three peaks ($m/z = 1838.069$, 1894.183 and 1949.222, respectively) corresponding to the exo olefin difunctional PIB (structure shown in Table 1) containing 27, 28, and 29 IB units, respectively (Fig. 5). This suggests that the reaction between living PIB and allyltrimethylsilane (ATMS) was incomplete. The resulting PIB-dichloride may eliminate HCl upon standing or under MALDI conditions as reported before. [24] The second series of three peaks ($m/z = 2066.584$,

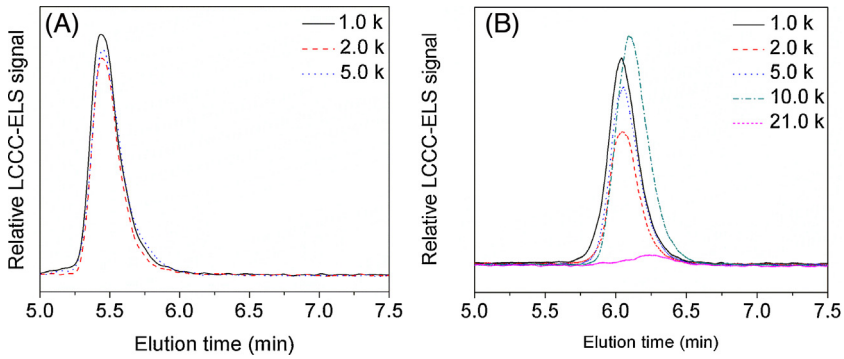


Fig. 6. LCCC chromatograms of PIB-diol (A) and PIB-diallyl (B) samples of varying MW under critical condition using ELS detector: THF/methanol 80.5/19.5 (w/w).

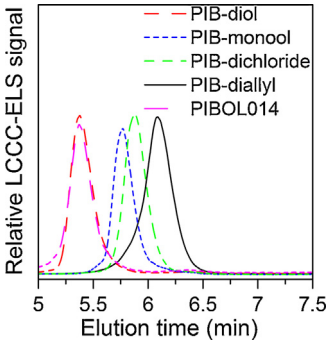


Fig. 7. LCCC chromatograms of PIB samples of varying functionality under critical condition using ELS detector: THF/methanol 80.5/19.5 (w/w).

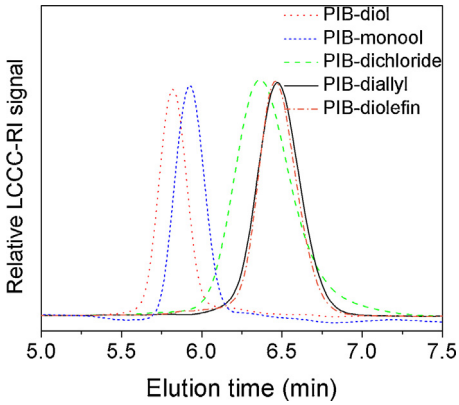


Fig. 9. LCCC chromatograms of PIB samples under critical condition using RI detector: THF/methanol 80.5/19.5 (w/w).

2122.639 and 2178.729) in the enlarged spectrum of f_2 corresponds to the PIB-monool (Table 1) containing 30, 31, and 32 IB units that arise from monoallyl functional PIB. The three main peaks (m/z = 3080.751, 3136.809 and 3192.832) of f_3 corresponds to the high MW PIB-diol containing 47, 48 and 49 IB units, respectively and the three peaks (m/z = 1845.102, 1901.151 and 1957.248) of f_4 corresponds to the PIB-diol containing 25, 26 and 27 IB units, respectively (Fig. 5). It is interesting to note is that f_3 contains polymers of higher MW than f_4, i.e., high MW polymers elute out earlier than lower MW polymers. This can be easily explained by considering the opposite thermodynamic characteristics associated with LC retention of hydroxyl end groups and PIB (alkyl) chain. The earlier elution of the polymer with longer alkyl chain is governed by the entropy-driven effects for PIB chain, while positive enthalpic interaction of the hydroxyl end groups is responsible for the retention. Furthermore, the polar hydroxyl groups have less impact on the overall polarity in the context of a higher molecular weight polymer, and therefore exhibit weaker interaction with the column in comparison with shorter chains of the low molecular weight polymer bearing two hydroxyl groups. Additionally, the gradient towards thermodynamically better solvent causes a slight shift to stronger SEC behavior. Such an inverse MW dependent retention behavior has been observed in HPLC systems showing hydrophobic interaction between nonpolar stationary phase and amphiphilic polymers in aqueous mobile phase, [25] but it is new to find the similar behavior in a non-aqueous mobile phase HPLC system. A closer inspection of the MALDI-TOF MS spectrum of f_2 revealed an unidentified series (highlighted in red in Fig. 5) with m/z values 2055.544, 2111.597 and 2167.681. At this point, we don't know the origin of these additional peaks. While the composition of the sample in Fig. 3 could be calculated by peak integration, the results are approximate since the ELS detector is a non-linear detector (see below).

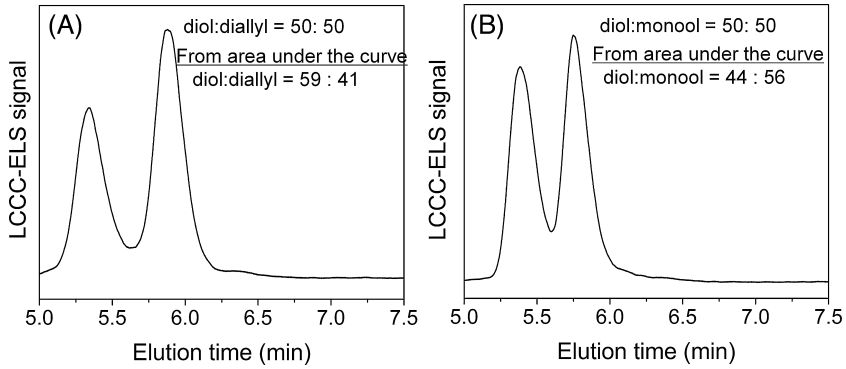


Fig. 8. LCCC chromatograms of mixture of PIB-diol/PIB-diallyl (A) and PIB-diol/PIB-monool (B) under critical condition using ELS detector: THF/methanol 80.5/19.5 (w/w).

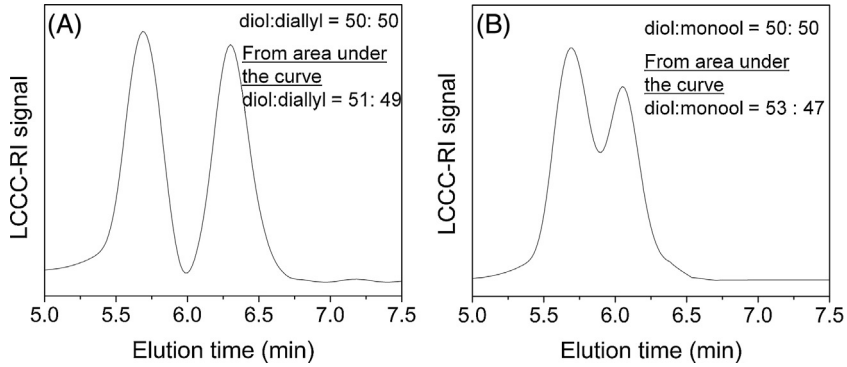


Fig. 10. LCCC chromatograms of mixture of PIB-diol/PIB-diallyl (A) and PIB-diol/PIB-monool (B) under critical condition using RI detector: THF/methanol 80.5/19.5 (w/w).

3.3. LCCC experiments: determination of critical conditions of adsorption

To determine the critical condition of adsorption, PIB-diol samples of different MW (approximately 1000, 2000 and 5000) and PIB-diallyl samples of different MW (approximately 1000, 2000, 5000, 10,000 and 21,000) were analyzed at different ratios of THF and methanol. According to Fig. 6 in THF/methanol 80.5/19.5 (*w*/*w*) PIB samples with identical chemical functionality eluted at the same position, independently of their MW. The same THF/methanol 80.5/19.5 (*w*/*w*) composition has already been employed by Mueller and coworkers for the LCCC analysis of PIB block copolymers. [26] However, this solvent composition is not suitable for high MW PIB as suggested by the very low sample recovery observed with M_n = 20900 due to precipitation of the polymer.

3.4. LCCC experiments: analysis at critical conditions using ELS detector

LCCC chromatograms of PIB samples of different functionality but similar MW (approximately 2000) under critical condition using ELS detector are shown as Fig. 7. PIB samples of the same MW (namely PIB-monool, PIB-diol, PIB-diallyl and PIB-dichloride) eluted at different elution time, depending on their chemical functionality under LCCC condition. As expected bifunctional PIB-diol eluted first, well separated from PIB-monool followed by non-functional PIB-diallyl (Fig. 7). However, when mixed PIB-diol and PIB-diallyl was injected, the ratio of PIB-diol and PIB-diallyl calculated from the area under the peak (Fig. 8) was slightly different from the actual ratio in the sample. Similarly, PIB-monool mixed with PIB-diol yielded two distinct peaks but the calculated ratio from the peak areas differed from the original composition. Thus, the results obtained using ELS detector as a concentration detector in the gradient HPLC method above or in the LCCC method may not be accurate.

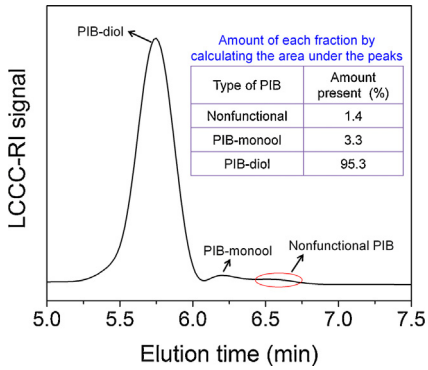


Fig. 11. LCCC chromatograms of PIBOL014 under critical condition using RI detector: THF/methanol 80.5/19.5 (w/w).

3.5. LCCC experiments: analysis at critical conditions using RI detector

Fig. 9 depicts LCCC chromatograms of PIB samples of different functionality but similar MW (approximately 2000) under critical condition using RI detector as the concentration detector. PIB samples (namely PIB-monool, PIB-diol, PIB-diallyl and PIB-dichloride) of the same MW eluted at different elution time depending on their chemical functionality under LCCC condition. PIB-diol eluted earlier compared to PIB-monool, while PIB-diallyl and exo olefin difunctional PIB (PIB-diolefin) eluted at a higher elution time compared to PIB-monool. Now, to check whether the LCCC-RI signal is proportional with the concentration of the PIB species or not, we mixed PIB-diol/PIB-diallyl (Fig. 10A) and PIB-diol/PIB-monool (Fig. 10B) at 50:50 ratio and injected them into the LCCC-RI system. Results revealed that the LCCC-RI signal is proportional to the concentration of the PIB species present in the mixture (Fig. 10). Thus, RI detector emerged as a better concentration detector under LCCC condition. The shift of the elution volume is due to the difference in kind of detection. RI detector has a large internal volume due to tubing inside the RI detector that the sample has to travel through. Therefore the RI detector will have a longer delay in retention time compared to the ELS detector.

The LCCC chromatogram of PIBOL014 revealed three peaks (Fig. 11) with elution times correlating to PIB-diol, PIB-monool and PIB without —OH functionality. Since according to MALDI-TOF MS, PIBOL014 does not contain PIB-diallyl this peak is attributed to PIB-diolefin, which may have formed from PIB-dichloride by elimination of HCl upon standing or during polymerization/end functionalization.

4. Conclusions

We have shown that the chemical homogeneity of PIB-diol can be determined by the combination of LCCC and MALDI-TOF MS. The ELS detector signal intensity is not strictly proportional with concentration. However, using LCCC with an RI detector the components PIB-diol, PIB-monool, and PIB without —OH functionality could be quantitatively determined. The overall $F_{n,OH}$ = 1.94 and $F_{n,OH}$ = 1.97, excluding PIB without —OH that does not affect the maximum number of chain extensions, suggests that PIBOL014 is of high quality. MALDI-TOF MS studies suggest that this can be further improved by extending the reaction time for allylation.

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Appendix A. Supplementary data

Supplementary data associated with this article can be found, in the online version, at http://dx.doi.org/10.1016/j.chroma.2014.12.022.

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